Ship Detection in Large-Scale SAR Images Via Spatial Shuffle-Group Enhance Attention

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***Abstract*— Ship target detection using large-scale synthetic aperture radar (SAR) images has important application in military and civilian fields. However, ship targets are difficult to distinguish from the surrounding background and many false alarms can occur due to the influence of land area. False alarms always occur with ship target detection because most of the area in large-scale SAR images are treated as background and clutter, and the ship targets are considered unevenly distributing small targets. To address these issues, a ship detection method in large-scale SAR images via CenterNet is proposed in this article. As an anchor-free method, CenterNet defines the target as a point, and the center point of the target is located through key point estimation, which can effectively avoid the missing detection of small targets. At the same time, the spatial shuffle-group enhance (SSE) attention module is introduced into CenterNet. Through SSE, the stronger semantic features are extracted while suppressing some noise to reduce false positives caused by inshore and inland interferences. The experiments on the public SAR-ship-data set show that the proposed method can detect all targets without missed detection with dense-docking targets. For the ship targets in large-scale SAR images from Sentinel 1, the proposed method can also detect targets near the shore and in the sea with high accuracy, which outperforms the methods like faster R-convolutional neural network (CNN), single-shot multibox detector (SSD), you only look once (YOLO), feature pyramid network (FPN), and their variations.**

***Index Terms*— CenterNet, large-scale scene, ship detection, spatial shuffle-group enhance, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has the unique advantages in that it is not limited by time of day or any type of inclement weather, has multiview imaging, and possesses certain penetration capabilities. Ship detection is a very important application of SAR image data, which plays the significant role in military and civil fields. At present, many airborne and spaceborne SAR-imaging technologies have been employed internationally, resulting in an increase of high-quality and large-scale scenery data available [1], [2]. Thus, ship target detection in large-scale SAR images becomes a particularly important traditional task [3], [4].

Traditional SAR ship target detection algorithms are mainly divided into three stages: area selection, feature extraction, and classification [5]. The primary purpose of area selection is to locate the targets. Feature extraction is based on region selection to extract classification features that can accurately express the target area, which is the core of target detection. Commonly used features include scale invariant feature transform (SIFT) [6] and histogram of oriented gradient (HOG) [7]. These methods must use gray value, scale size, or other characters to manually design features and classifiers. These features perform poorly for ship detection in inshore areas and it is difficult to rule out false positives, such as icebergs and islands. At the same time, remodeling is required for different scenes, which causes low detection efficiency and high false alarm rate [8]. In addition, speckle noise and motion blur in SAR images may cause unnecessary differences between ships, which make it difficult for traditional SAR ship detection [9]. At present, the constant false alarm rate (CFAR) detection algorithm is the most widely used and effective detection method in the theoretical research of SAR image ship detection. However, the CFAR algorithm is greatly affected by the background statistical area. Generally speaking, increasing the background statistical area increases the accuracy of the description, but this may result in larger background clutter changes and increase the false alarms rates [10]. Thus, it is difficult to solve the issues of ship target detection using the traditional methods for large-scale SAR images [11].

With the development of neural networks, target detection technology based on deep learning has become the mainstream direction in the field of target detection. An increasing number of target detection methods based on deep learning have been proposed, through which higher accuracy can be obtained. Especially in recent years, the end-to-end networks have been proposed, in which design features do not need to be manually designed, and in turn, enhances the generalization ability [12]. The emergence of AlexNet [13] has epoch-making significance, making it possible for convolutional neural networks (CNNs) to go deeper. As a result more in-depth research on target detection networks have been conducted to produce new algorithms. These algorithms mainly include: two-stage detection algorithms represented by faster R-CNN [14] and single-stage detection algorithms represented by you only look once (YOLO) [15] and single

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shot multibox detector (SSD) [16]. To solve the shortcomings of fewer shallow features of the target on the final feature map, a feature fusion method feature pyramid network (FPN) structure is proposed [17].

Ship detection in SAR images based on CNNs has also had great breakthroughs [18], [19]. For instance, Li *et al*. [8] proposed an improved faster R-CNN method. In [1], a squeeze and excitation rank faster R-CNN method was proposed for ship detection in SAR images. Kang *et al*. [20] put forward a contextual region-based method for SAR ship detection. Jiao *et al*. [21] densely connected multiscale neural network to solve the problem of multiscale and multiscene SAR ship detection. All above methods are based on anchors. The matching mechanism of the anchor makes the frequency of extreme scales (large and small targets) to be matched lower than that of other moderately sized targets. But the size of the ship targets varies widely in large-scale SAR images. Therefore, the anchor-based method is not suitable for ship target detection in large-scale SAR images.

Recently, researchers have focused on the research of anchor free methods, such as CornetNet [22] and ExtremeNet [23]. CornerNet completely abandons the anchor concept and uses the method of detecting points to identify targets for the first time. CornerNet detects the upper-left and lower-right corners of the bounding box to determine the position and size of the target. However, the semantic information in the top left and bottom right is relatively weak and difficult to detect. At the same time, two points that belong to the same target need to be combined with postprocessing. ExtremeNet uses the best keypoint estimation framework to find extreme points by predicting four multipeak heatmaps for each target class. In addition, ExtremeNet uses the center heatmap of each class to predict the target center. However, the speed is slow and more points need to be detected and the detection effect is not significantly improved. Too much attention to the edge can easily cause missed inspections and false alarms. CenterNet [24] is improved on the above basis, and the detection is based on the center point of the target. CenterNet only needs to extract a center point for each object without grouping or postprocessing. Therefore, it can effectively reduce missed detections in large-scale SAR images.

In large-scale SAR images, the proportion of the target area is usually very small, which means that ship detection on large-scale SAR images is considered as small object detection. When mapped to the final feature map, small objects have little information for location refinement and classification, which naturally reduces detection performance. Therefore, the CenterNet-based target detection method is proposed in this article and is used to solve the problem of detecting ships docking side by side on the shore. However, there are still many false alarms in the near shore and inland areas.

Generally, ship detection in large-scale SAR images has many disturbances, such as land, islets, and sea clutter, as shown in Fig. 1. To reduce possible false alarms, attention mechanisms have been introduced to target detection network. Attention mechanisms have made important breakthroughs in image interpretation, natural language processing, and other fields. Attention mechanisms have also been helpful in the improvement of network model performance in recent years [25]. It is also functions similar to the perception mechanism of the human brain and human eye by focusing on local information mechanisms [26]. The results are usually presented in the form of probability graphs or probability eigenvectors. In principle, attention models are mainly divided into three types: spatial attention model, channel attention model, and spatial and channel mixed attention model. SENet [27] proposes the squeeze and excite (SE) module, which essentially performs attention operations on channels. But in addition to the channel, another important dimension in the convolutional feature map needs to be considered: space. Channels and spaces mix attention to get more detailed semantic features.

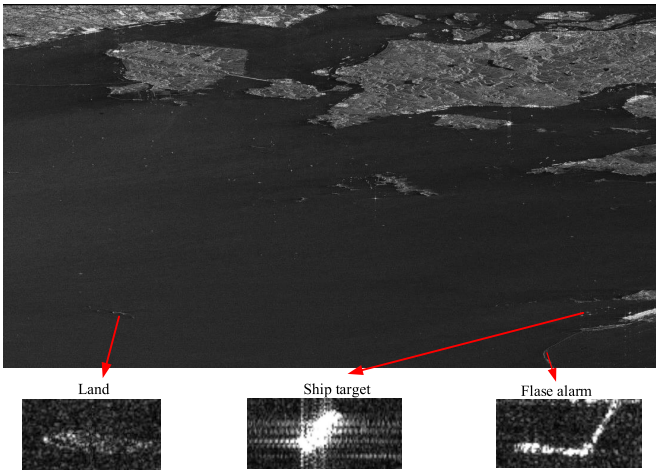


Fig. 1. Large-scale SAR image with many disturbances.

Based on above analysis, the spatial shuffle-group enhance (SSE) attention module is proposed to get semantic features that better represent target features. SSE divides the feature map into G groups along the channel dimension. Then the channels shuffle to enhance the relationship between the groups. The attention model SSE is combined with the anchor-free CenterNet network. In this way, SSE enables feature maps with similar semantics to better extract the regional features of the targets. Experiments show that the method proposed in this article has better detection performance and high accuracy on ship detection in large-scale SAR images.

The rest of this article is organized as follows: Section II introduces the proposed method based on CenterNet [24]; In Section III, the details of experiments on the data set named SAR-ship-data set and the analysis of the results are exhibited; finally, the conclusion is presented in Section IV.

II. Proposed Method

The overall process framework shown in Fig. 2 mainly consists of rough land and sea classification and ship detection by CenterNet with SSE attention module. In this section, the overall process and key technical points of the proposed method are introduced, and then each branch of the proposed method is described in detail.

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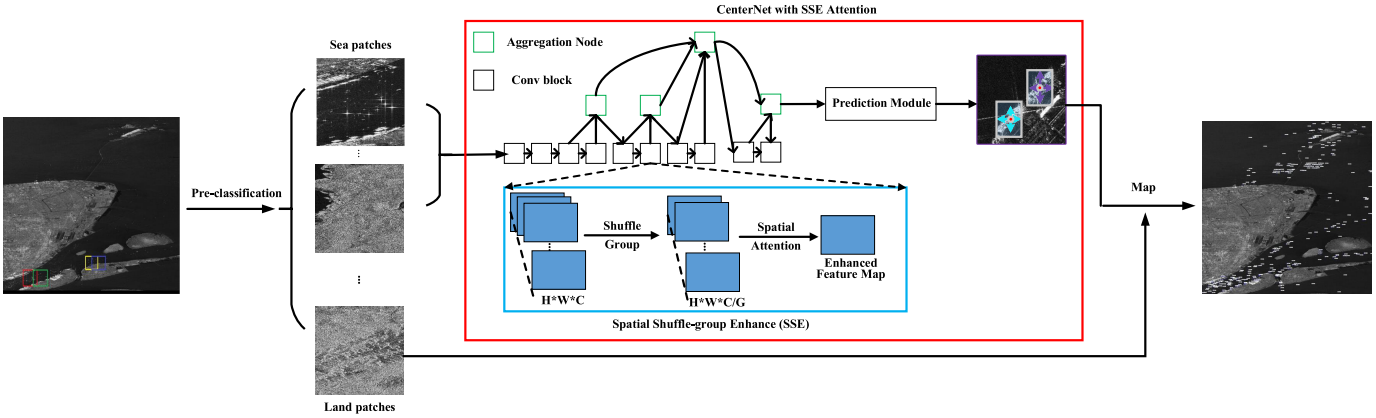


Fig. 2. Overall framework of the proposed method for ship detection.

A. Ship Detection Framework in Large-Scale Images

As SAR imaging technology has developed, the SAR image resolution has improved gradually, and data sets have become gradually more enriched. In large-scale SAR images, the proportion of ship targets in the entire image is very small. At the same time, the image size is large (more than one million pixels) and cannot be directly input into existing deep neural networks. Therefore, before sending the SAR image of the large scene into the neural network, a sliding window is needed to divide the SAR image of the large scene into small slices. Sliding window is a preprocessing process in the ship target detection process of large scenes. A fixed window size (400×400) without overlapping is used for sliding windows. However, pixel-by-pixel detection produces more false alarms, which wastes computing resources. A rough classification method of sea and land is proposed to reduce the waste of computing resources and to reduce the false alarms inland before detection.

In large-scale SAR images, the size of the ship targets varies greatly. Especially the detection of small ship targets is difficult. Researchers have proposed many detection frameworks for small targets [28], [29]. These methods mainly focus on implementing multiscale frameworks or using reinforcement methods to make the network pay more attention to small targets. Compared with other target detection frameworks, CenterNet is an anchor-free detection network based on center points [24], which greatly reduces the amount of calculation. And the down sampling factor of CenterNet's output resolution is 4, which is relatively small when compared with other target detection frameworks (Mask-Rcnn is at least 16 and SSD is at least 16). Thus, CenterNet is chosen as the target detection network framework in this article.

Although CenterNet fuses shallow features with deep features, it cannot discriminate well for some false alarms in large-scale SAR images, such as land and islands. Thus, obtaining more advanced semantic features is essential to eliminate false alarms. Feature learning and grouping in convolutional networks have been primary areas of research in recent years. Numerous studies have previously studied the spatial components of CNNs. They all try to enhance the representation ability of CNNs by improving the quality of the spatial coding of the entire feature level.

In recent years, grouping channels has been applied to reduce the amount of computation. Inspired by spatial group-wise enhance (SGE) module [30], the SSE attention module proposed in this article is used to simultaneously integrate attention into channels and spaces. The channels are grouped to reduce the amount of calculation, and then are shuffled to enhance the connection between the groups. In the space of the group, global features and critical features are combined to obtain stronger semantic information. The important information of the feature makes the feature more expressive of the target characteristic. Thus, it can extract higher level semantic features, and reduce false alarms.

When processing on a large scene image, the large scene image is cropped to a small scale (e.g., 400×400) without overlapping to handle the large input size and work within the computer's limited memory. The patches of large-scale SAR images are roughly classified by sea and land. The patches that contain only land (such as the last patch in Fig. 2) are screened out and not sent to the heavy detector. The rest patches are sent to the detector to obtain the detection results of the ship target. Finally, all the patches are mapped back to the original image through the positional information to obtain the final detection results.

B. CenterNet

Most current target detectors list potential target locations and classify each target. This type of methods rely heavily on the design of hyperparameters such as anchors and thresholds, which are computationally intensive. CenterNet models objects as a single point, finds the center point by keypoint estimation, and uses the image information of the center point to get the size and position of the targets. There is no need to set anchors in advance, which greatly reduces the amount of network parameters and calculations.

CenterNet has many advantages for being applied to SAR ship detection. First, it has no artificial thresholds for foreground and background classification, which reduces requirements for positive and negative samples of the data set. At the same time, the difficulty of detecting small targets due to the anchor matching mechanism is reduced. What is more, the center point can be regarded as a single agnostic

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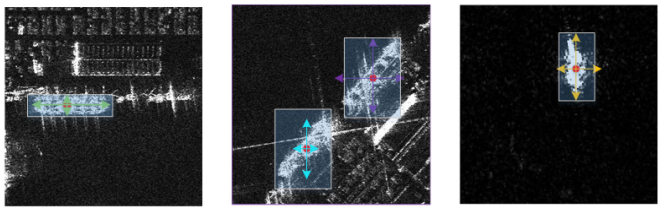


Fig. 3. Model an object as the center point of its bounding box. The bounding box size and other object properties are inferred from the keypoint feature at the center.

shape anchor, where only local peaks are extracted in the keypoint heatmap, and nonmaximum suppression (NMS) is not needed. Detection speed is greatly accelerated without NMS, since NMS is extremely time consuming. At the same time, the anchor-based target detection method requires NMS to obtain the final detection result after obtaining candidate regions. In the process, densely docked ship targets are likely to be suppressed. As a result, densely docked ship targets are missed. CenterNet is no need for NMS, and only returns a bounding box based on the center point of the target, the detection of densely docked ship targets does not affect each other.

In addition, compared with traditional target detection (output stride of 16), CenterNet uses a higher output resolution (output stride of 4), which can better retain the characteristics of small targets. CenterNet uses a full convolutional network with encoding and decoding. Up sampling uses transposed convolution, which can better restore semantic information and position information of the images. At the same time, CenterNet is end-to-end differentiable, simple and fast.

In large-scale SAR images, the targets occupy a small proportion of the entire image, so better feature extraction is particularly important. Therefore, CenterNet-based object detection method is adopted for ship detection. CenterNet inputs images to the full convolutional network to generate a heatmap, and then screens to get the final target center point. Image features at the peak can predict height and width of the target bounding box.

The specific implementation principle is shown in Fig. 3. First, assume $I \in R^{W\times H\times 3}$ is a patch with input width H and height W. Obeying $\hat{Y} \in [0,1]^{(W/R)\times(H/R)\times C}$ distribution, where R is the output stride size and C is the number of keypoint categories, keypoint heatmap is obtained. If $\hat{Y}=1$, it corresponds to a detected key point; if $\hat{Y}=0$, this is the background. If (x_1, y_1, x_2, y_2) is the bounding box of the target, its center point is at $p = ((x_1+x_2)/2, (y_1+y_2)/2)$. The other information of the target is obtained from the image information of the key points. All the center points are predicted by $\hat{Y}$ and then regressed to get the target size. The value at the key point is used as its confidence information, and regression at its position is used to get the bounding box size. The position coordinates are

$$\left(\hat{x}_i+\delta\hat{x}_i-\frac{\hat{w}_i}{2}, \hat{y}_i+\delta\hat{y}_i-\frac{\hat{h}_i}{2}, \hat{x}_i+\delta\hat{x}_i+\frac{\hat{w}_i}{2}, \hat{y}_i+\delta\hat{y}_i+\frac{\hat{h}_i}{2}\right) \tag{1}$$

where $(\delta\hat{x}_i, \delta\hat{y}_i) = \hat{O}_{\hat{x}_i,\hat{y}_i}$ is the offset prediction in the down sampling process. $(\hat{w}_i, \hat{h}_i) = \hat{S}_{\hat{x}_i,\hat{y}_i}$ is the prediction of the target size.

However, CenterNet only detects the center point and obtains minimum target information, which results in more false positives for ship target detection in SAR images. To solve this problem, SSE is applied to the network so that the network can focus on more advanced semantic information.

C. Spatial Shuffle-Group Enhance Attention

In [30], a SGE module was proposed, which is the first-generation source to use the similarity between local and global as an attention mask. Furthermore, SGE has good interpretability for the enhancement of semantic representation. The article believes that a complete feature is composed of many subfeatures, and these subfeatures are distributed in the form of groups in the features of each layer. The SGE attention module obtains the importance of each subfeature by generating an attention factor for each spatial position in each group, where each group can learn and suppress noise in a targeted manner. This attention factor is determined only by the similarity between global and local features within each group. SGE is a lightweight attention module that obtains high-level semantics, which can significantly improve target detection performance. However, these subfeatures were processed in the same way, and they were affected by background noise, which lead to incorrect classification and positioning results. When multiple groups of convolutions are superimposed together, the output of a certain channel is only derived from a small number of input channels, thereby reducing the information flow and information representation capabilities between channel groups.

In [31], an effective CNN named Shufflenet for mobile devices was proposed, which is mainly used for compression and acceleration of deep networks. Modern CNNs contain multiple repetitive modules. As the number of network layers deepens and the number of channels increases sharply, the convolution kernel of the convolutional layer increases, and the spatial dimension decreases accordingly. At the same time, the feature map becomes decreases with the convolution pooling operation. As a result, channels are becoming more and more important in deep networks. Group convolution can solve the limitation between limited channels. By ensuring that each convolution operation is performed only on the corresponding input channel group, group convolution can significantly reduce calculation loss. If group convolution is allowed to receive different input data, the input and output channels will become fully correlated.

Mainly inspired by the above research, the SSE module is proposed in this article. The idea of channel shuffle in model compression is applied to our attention module. The overall framework of the proposed attention module (SSE) is shown in Fig. 4. SSE divides the C channels and $H\times W$ convolutional feature map into G groups along the channel dimension. When the group shuffle is applied to increase the correlation between the groups, SSE shuffles the output channels of the previous layer, divides them into G groups again, and outputs to the next layer, as shown in Fig. 5. SSE can aggregate more relevant

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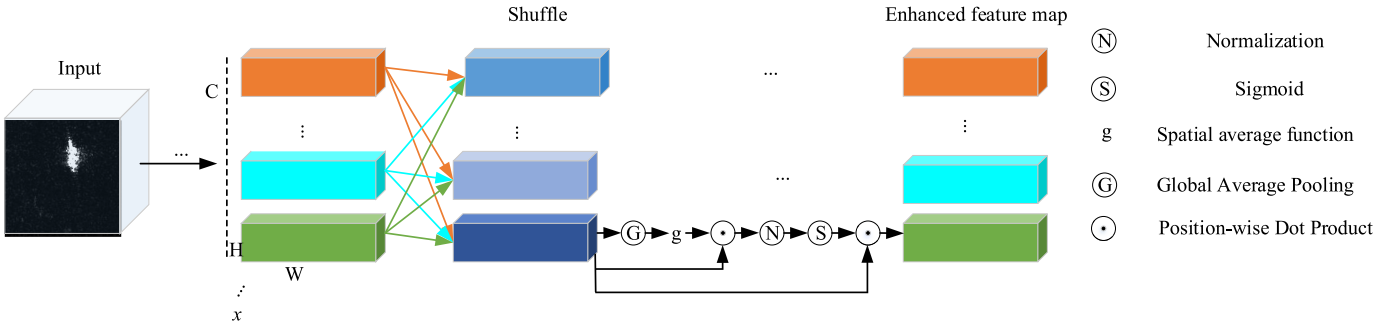


Fig. 4. Structure of SSE module. Semantically enhanced feature maps are obtained through this attention module.

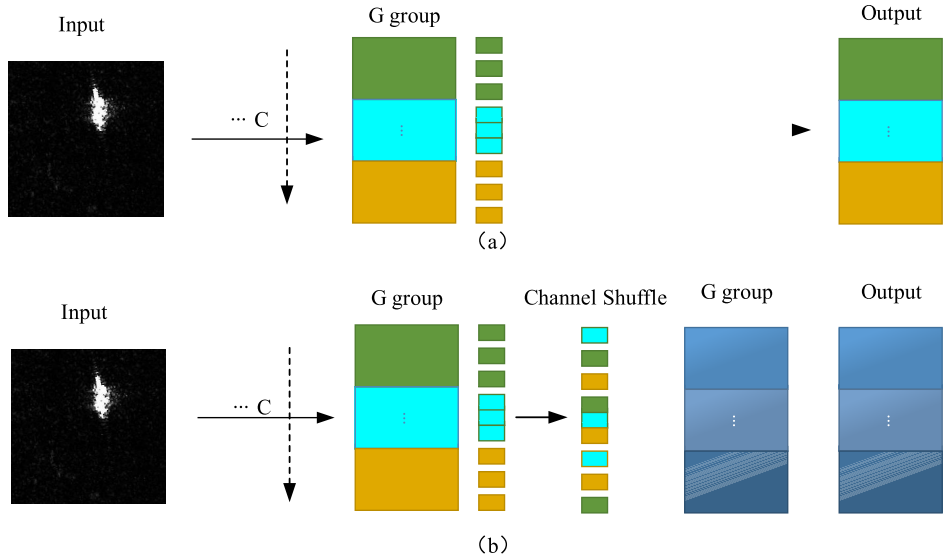


Fig. 5. Channel shuffle operation in convolutional layer. (a) Input and output channels are fully related. (b) Each output channel only relates to the input channels within the group.

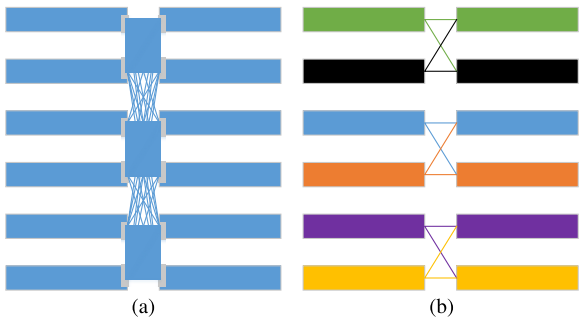


Fig. 6. Standard convolution and group convolution. (a) Normal convolutional layer. (b) Convolutional layer with three groups.

features between channels to improve detection performance, and some false alarms can be eliminated. Meanwhile, SSE generates an attention factor at each spatial location within each group to learn more advanced semantic information, which produces enhanced feature maps [30].

In deep networks, channels become more important. Group convolution can reduce the amount of calculations significantly [23], and its schematic is shown in Fig. 6. It can be seen from Fig. 6 that each group learns different features by a fewer parameters. Group convolution is to group input feature maps, and then convolve each group separately. Suppose the size of the input feature map is $C \times H \times W$, and the number of output feature maps is N. If channels are divided into G groups, the size of each convolution kernel is $(C/G) \times K \times K$, and the number of convolution kernels is still N. The convolution kernel is only convolved with the feature maps of the same group. The total parameter amount is $N \times (C/G) \times K \times K$, and the total parameter amount is reduced to the original $(1/G)$. Group convolution can be regarded as sparsely structured. The size of each convolution kernel changes from $C \times K \times K$ to $(C/G) \times K \times K$. The remaining $(C - (C/G)) \times K \times K$ parameters can be regarded as 0, which can sometimes result in even better results while still reducing the amount of parameters.

Channel shuffle operation can be completed by the following steps [31]: first, divide the channels of a convolutional layer into G groups, where each group has $n = (C/G)$ channels. Then, reshape the output channel dimension into (G, n) and transpose into (n, G). Next, flatten and redivide into G groups.

Due to noise, it is difficult for a CNN to obtain favorable feature distributions. To solve the problem, the overall information of the entire group is used to enhance the learning of semantic features in critical regions. The spatial enhance

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operation can be completed by the following steps: global statistical features are used to approximate the semantic vector g of this group of learning representations through the spatial average function

$$g = \frac{1}{m}\sum_{i=1}^{m} x_i \tag{2}$$

where $x = \{x_{1,\ldots,m}\}, x_i \in \Re^{(C/G)}, m = H \times W$ is the vector representation of the group at each position in space.
Next, a corresponding importance coefficient c_i is generated for each feature. The coefficient is obtained by a simple dot product, which measures the similarity between the global semantic feature g and the local feature x_i

$$c_i = g \cdot x_i. \tag{3}$$

At the same time, to prevent deviations in coefficients between various samples, c in space is normalized

$$\mu_c = \frac{1}{m}\sum_{j}^{m} c_j \tag{4}$$

$$\sigma_c = \frac{1}{m}\sum_{j}^{m} \left(c_j - \mu_c\right)^2 \tag{5}$$

$$\hat{c}_i = \frac{c_i - \mu_c}{\sigma_c + \varepsilon} \tag{6}$$

where ε is a constant added for numerical stability.

To ensure that the normalization inserted in the network can represent identity conversion, a set of parameters γ, β is introduced for each coefficient c_i to scale and transform the normalized value [32]. $a_i = \gamma \hat{c}_i + \beta$. Finally, to obtain the enhanced feature vector $\hat{x}_i$, the original x_i is scaled spatially by the generated importance coefficient a_i through the sigmoid function threshold

$$\hat{x}_i = x_i \cdot \sigma(a_i). \tag{7}$$

All enhanced features make up the final feature map. The enhanced feature map can better exclude the interference of the coast on ship target detection. This greatly improves the accuracy of detection and reduces false alarms on the shore. Therefore, CenterNet with SSE solves the problem of detecting ships docking side by side at the shore.

The SSE module can be embedded in existing network structures the same as the SE module. The SSE module is placed after the last BatchNorm layer of each bottleneck of CenterNet, as shown in Fig. 7.

III. Experiments

To evaluate the proposed method, experiments are implemented on public data set named SAR-ship-data set [33]. First, the implementations such as data set, evaluation criteria, and detailed experimental settings will be introduced. Next, the role of SSE is mainly verified, and the current ship target detection methods are compared to verify the effectiveness of the proposed method in this article. At last, the effectiveness of the proposed method in large-scale scenes is verified.

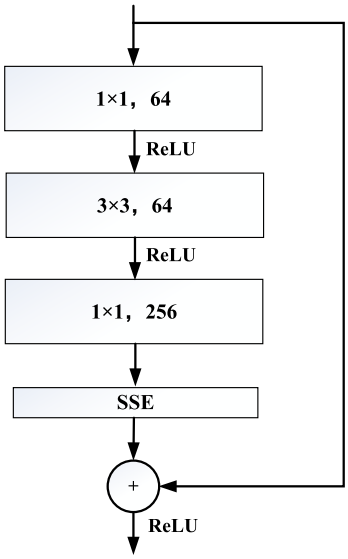


Fig. 7. Position of SSE in the bottleneck module.

A. Implementations

1) Data Sets: SAR-ship-data set is a multisource, multimode SAR ship data set published by the Chinese Academy of Sciences. It contains 102 Gaofen-3 images and 108 Sentinel-1 images that are used to create 59 535 ships in 43 819 ship chips [33]. The information of the original SAR image is more abundant, and the resolution, incident angle, polarization method, and imaging mode are all very different. This information provides beneficial conditions for ship detection in SAR images based on deep learning.

For ship target detection in large scene SAR images, some typical slices from Sentinel 1 are applied, which have different polarization methods, vertical transmission and vertical reception (VV) and vertical transmission and horizontal reception (VH). The original pixel values of the three large-scale SAR images shown in the article are 12 075 × 8351, 4417 × 7749, and 8560 × 8297.

2) Evaluation Metric: To evaluate the detection performance of the network, AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l are used as the metrics following the evaluation protocol on the COCO data set. Average precision (AP) is the area under the precision–recall curve. Mean average precision (mAP) is the average of multiple categories of AP. mAP is the most important one in the target detection algorithm. There is only one category of ships in this article, so mAP = AP. Generally speaking, the better the classifier, the higher the AP value. In this article, AP is average precision at Intersection over Union (IoU) = 0.50:0.05:0.95 (primary challenge metric), and AP_{50} is AP at IoU = 0.5 (PASCAL Visual Object Classes (VOC) metric), Similarly, AP_{75} is AP at IoU = 0.75 (strict metric). AP_s, AP_m, and AP_l represent AP for small objects, medium objects, and large objects, respectively [34]. The most important metric is AP_{50}. The precision and recall are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

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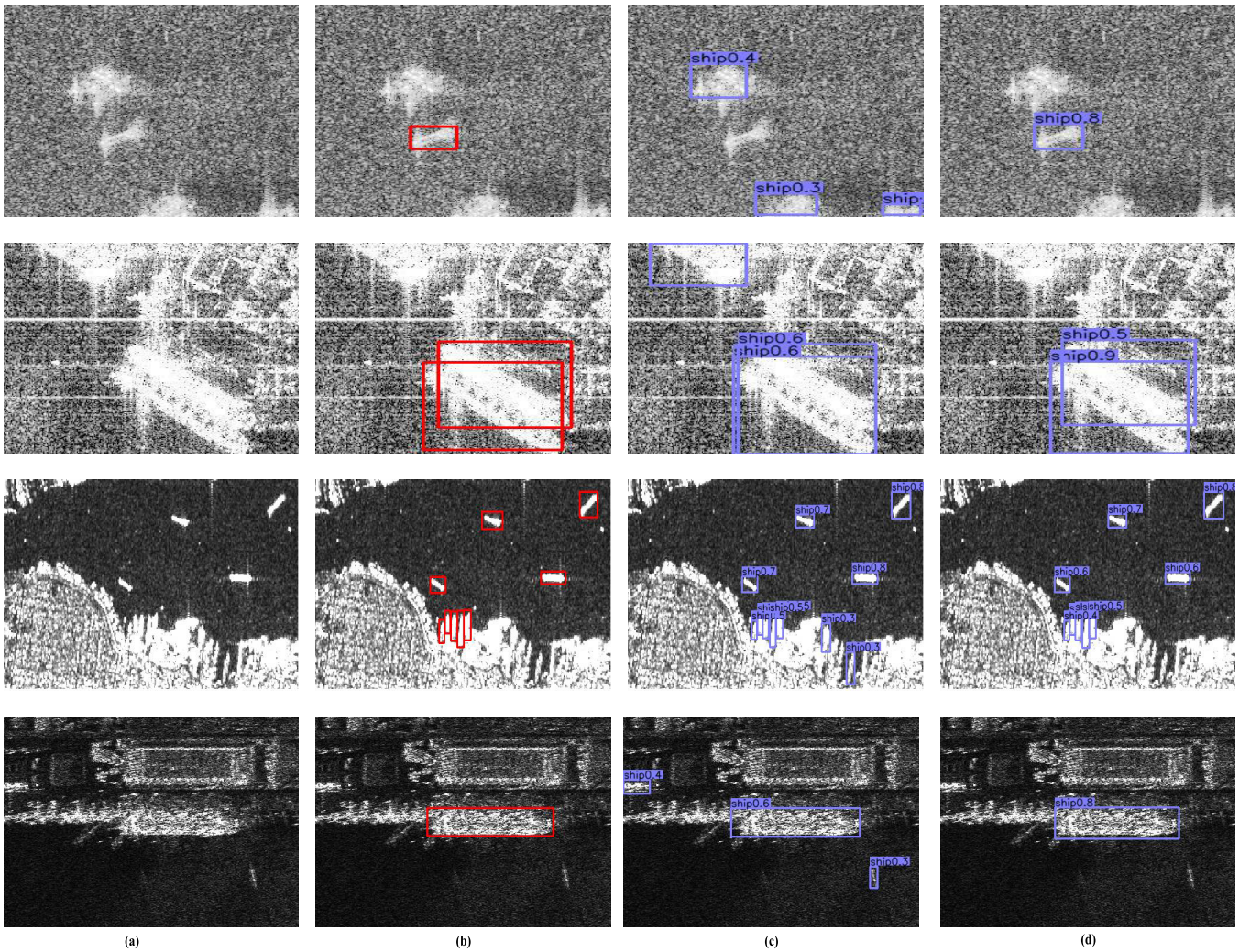


Fig. 8. Effect of SSE module. (a) is the original images. (b) is the ground truth. (c) is the detection results of CenterNet without SSE. (d) is the detection results of CenterNet with SSE.

where TP (true positives) is the number of correctly detected ships and FP (false positives) is the number of ships incorrectly classified as positive, and FN (false negative) is the number of ships incorrectly classified as negative.

AP is defined as

$$\mathrm{AP} = \int_0^1 p(r)dr \tag{10}$$

where p represents precision, r represents recall, and p is a function that takes r as a parameter, which is equal to taking the area under the curve. In addition, the training time is applied to the comparison to show the time advantage of the method proposed in this article.

3) Settings: The proposed method is implemented on pytorch 0.4.1, CUDA 9.0 and CUDNN 7.1 with Intel Core i7-6700HQ CPU and a Nvidia Geforce GTX 1080Ti GPU. Data set is divided by random sampling, the ratio of training set, verification set, and test set is 8:1:1. There are 35 055 chips for training and 4382 chips for verification and test, respectively. The input resolution of 512 × 512 is applied, which will yield an output resolution of 128 × 128 for all the models. The operations of random flips, random scaling (between 0.6 and 1.3), crop and color dithering are used for data enhancement, and the Adam operation is used to optimize the overall goal. The learning rate is set to 1.25e−4, and the batch size is set to 16. The backbone of the CenterNet detection network uses full layer convolutional network deep layer aggregation (DLA) [35]. The DLA network is an image classification network with hierarchical skip connections. DLA is modified with bottleneck residual blocks and deformable convolution. The down sampling layer of DLA-34 is initialized using ImageNet pretraining, while the up sampling layer is initialized randomly.

B. Ship Detection in SAR-Ship-Data Set

1) Effect of SSE: In this section, the role of the attention module proposed in this article is verified. When training the networks, the same parameter settings are used for comparison. When detecting on large-scale images, slicing is necessary, so it is also valid to verify the model in small images. At present, the scope of comparison and reference in the detection of ship targets in large scenes is relatively small, and considering the problem of image display in this article, the effectiveness of the method is verified in the small images first. As shown in Table I, after adding the attention

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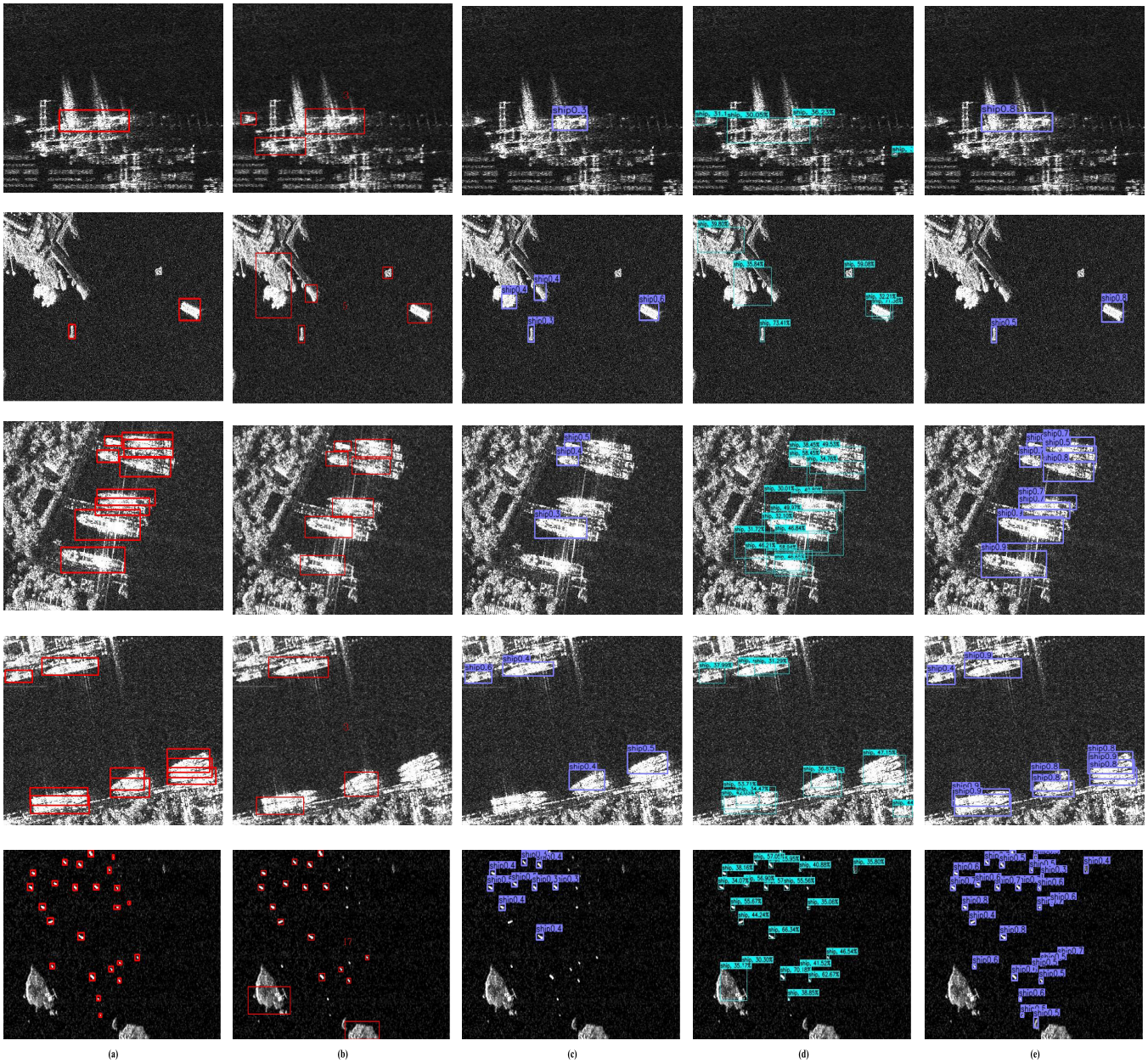


Fig. 9. Effect of the proposed module. (a) is the ground truth. (b) is the detection results with faster R-CNN. (c) is the detection results with SSD-300. (d) is the detection results with YOLOv3. (e) is the detection results with the proposed method.

module the AP_{50} reached 94.7%, which is 2.3% higher than before the addition. At the same time, the AP for small targets improved about 6%.

As shown in Fig. 8, [Fig. 8(a)] are the original images, [Fig. 8(b)] is the ground truth, and [Fig. 8(c) and (d)] represent test results with and without SSE, respectively. The red rectangle in [Fig. 8(b)] represents the real ship targets and the purple rectangle in [Fig. 8(c) and (d)] represents detected ship targets. As a note, a rectangular box in this article always represents a ship target. The ship targets in [Fig. 8(b)] is used as a reference, it can be seen from [Fig. 8(c)] that CenterNet has good detection results, but there are many false alarms. After adding SSE to the network structure, the detection of false positives decreases, as shown in [Fig. 8(d)] and the detection rate improves, as shown in Table I.

2) Comparison With Other Methods: To verify the performance of our method, a comparative analysis using different methods is conducted, followed by a detailed explanation of the experimental results. As illustrated in Fig. 9, it can be seen from the experiment results that, the method proposed in this article could effectively reduce false alarms when compared with other methods. At the same time, it has a better detection performance for dense-docking ships than other existing network models.

First, the method proposed in this article is compared with the ship target detection methods in the field of optics. As shown in Table II, the proposed method obtains the highest mAP. In Fig. 9, the comparison of the effects of several methods can be seen. [Fig. 9(a)] is the ground truth. [Fig. 9(b)] shows the results with method of faster R-CNN [33]. It can

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TABLE I
DETECTION PERFORMANCE OF CENTERNET WITH AND WITHOUT SSE

Method	$AP(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_s(\%)$	$AP_m(\%)$	$AP_l(\%)$
CenterNet	54.4	92.4	58.3	48.3	62.6	72.2
SSE-CenterNet	60.9	**94.7**	69.1	**54.7**	69.4	73.9

TABLE II
COMPARISON WITH THE TYPICAL DETECTION NETWORKS IN DEEP LEARNING

Method	Input Size	mAP(%)
Faster R-CNN	600×800	88.26
SSD-300	300×300	88.32
SSD-512	512×512	89.43
YOLOv3	416×416	85.2
the proposed method	512×512	**94.7**

TABLE III
COMPARISON WITH THE MAIN CURRENT SHIP DETECTION METHODS BASED ON FPN

Method	mAP(%)	times(ms)
Improved Faster R-CNN [37]	88.8	96.043
Improved FPN [36]	87.5	123.2
the proposed method	**94.7**	**18**

be seen from [Fig. 9(b)] that, there are many false alarms in the detection results, such as islands and land. At the same time, there are many missed detections for densely docked ships. [Fig. 9(c)] is the result with the SSD-300 method [33], in which there are many missed targets. [Fig. 9(d)] shows the results with the YOLOv3 method, which also shows many false alarms and has very low confidence. [Fig. 9(e)] shows the results using the proposed method. It can be seen from [Fig. 9(e)] that, the method proposed in this article could eliminate some false alarms and for densely docked vessels, and it has a better detection effect than all previous methods, without additional processing. These benefit from the feature map enhanced by the attention module SSE, and CenterNet gets the target bounding box by center point regression, without the need of NMS.

At the same time, it is worth mentioning that, small ship targets with fewer target pixels may be lost with the deepening of the network structure. The reason is that other CNN methods require a large amount of down sampling during the convolution process to reduce redundancy and reduce calculations. The method in this article only needs four times down sampling, which can detect more small ships.

The comparison experiments with the current SAR ship detection methods based on FPN were also implemented. An improved FPN structure was proposed in [36], with constructing the FPN through horizontal connections, and using the scale conversion layer to densely connect each feature map from top to bottom. Horizontal connections inject semantic information into high-resolution feature maps. In [37], the attention module convolutional block attention module (CBAM) was applied in FPN, and embedded in faster R-CNN. Both the methods in [36] and [37] are proposed for ship targets detection in SAR images, and the detection performance is shown in Table III. From Table III, it can be seen that, the method proposed in this article still has the highest mAP and the fastest detection speed compared with the main current ship detection methods.

C. Ship Detection in Large-Scale SAR Images

1) Preprocessing: With the improvement of SAR image resolution, people are more interested in solving the detection of ship targets in large-scale scenes. The proposed network integrates the prescreening and detection network into a unified framework, which takes into account that larger input images will increase computing time and memory consumption. The large-scale SAR images are cropped with 400 × 400 pixels, which solves the problem of insufficient computer memory. Then the full convolutional backbone structure is applied to the input to achieve powerful feature extraction, and the target is bounded by regression.

Preprocessing structure has two advantages.

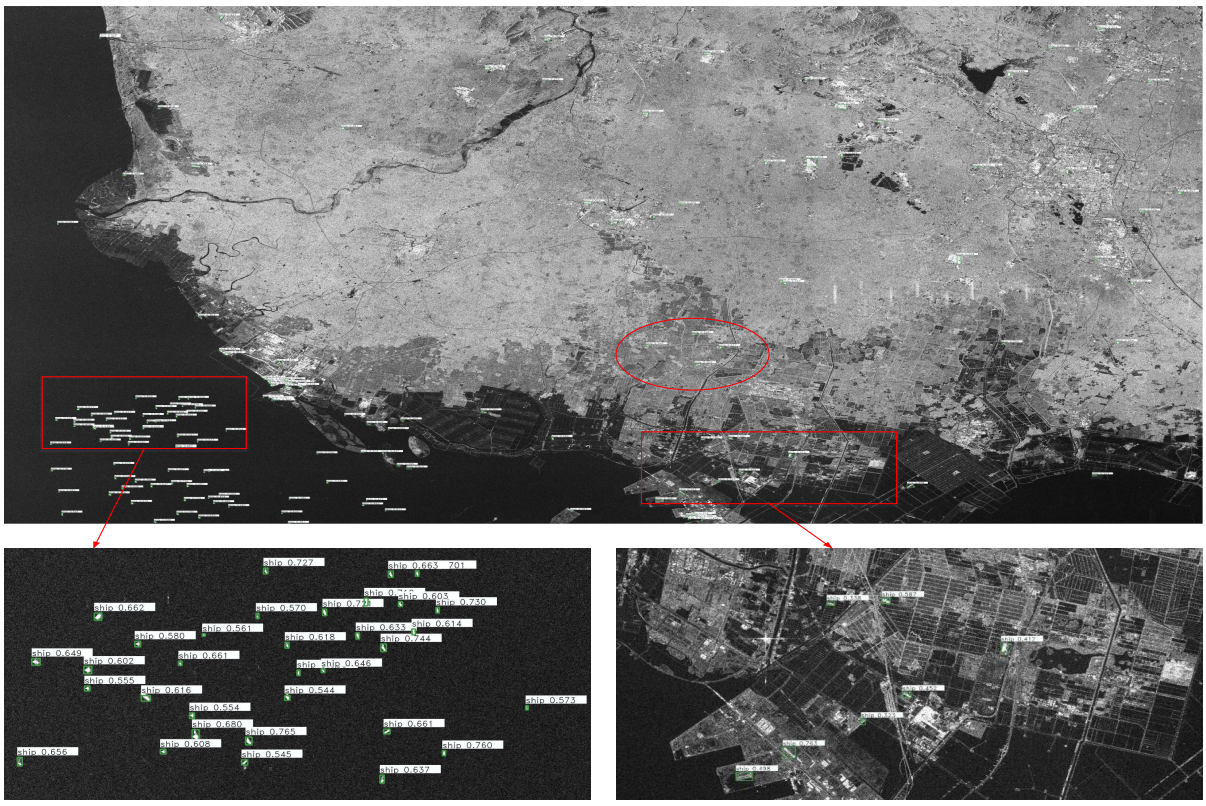
1) The preclassification step can screen out noncandidate regions without training and extensive calculations. The preclassification structure is shown in Fig. 2. Considering that there are no valid targets in many patches, these patches do not require subsequent heavy feature extraction and branch detection. As a result, the detection speed and accuracy was greatly improved. item The second advantage is that since a large and complex background area may cause more false positives, these false positives are first filtered out by the preclassification process.

For example, land construction areas have pixel characteristics similar to ships. Even if the preclassification is not correctly classified, subsequent tests can still correct the final test results. Although the preclassification method of sea and land can effectively reduce the interference of the land area to the ship target detection, it is dependent on the data set. If more images in the data set contain only sea areas, land and sea preclassification becomes redundant.

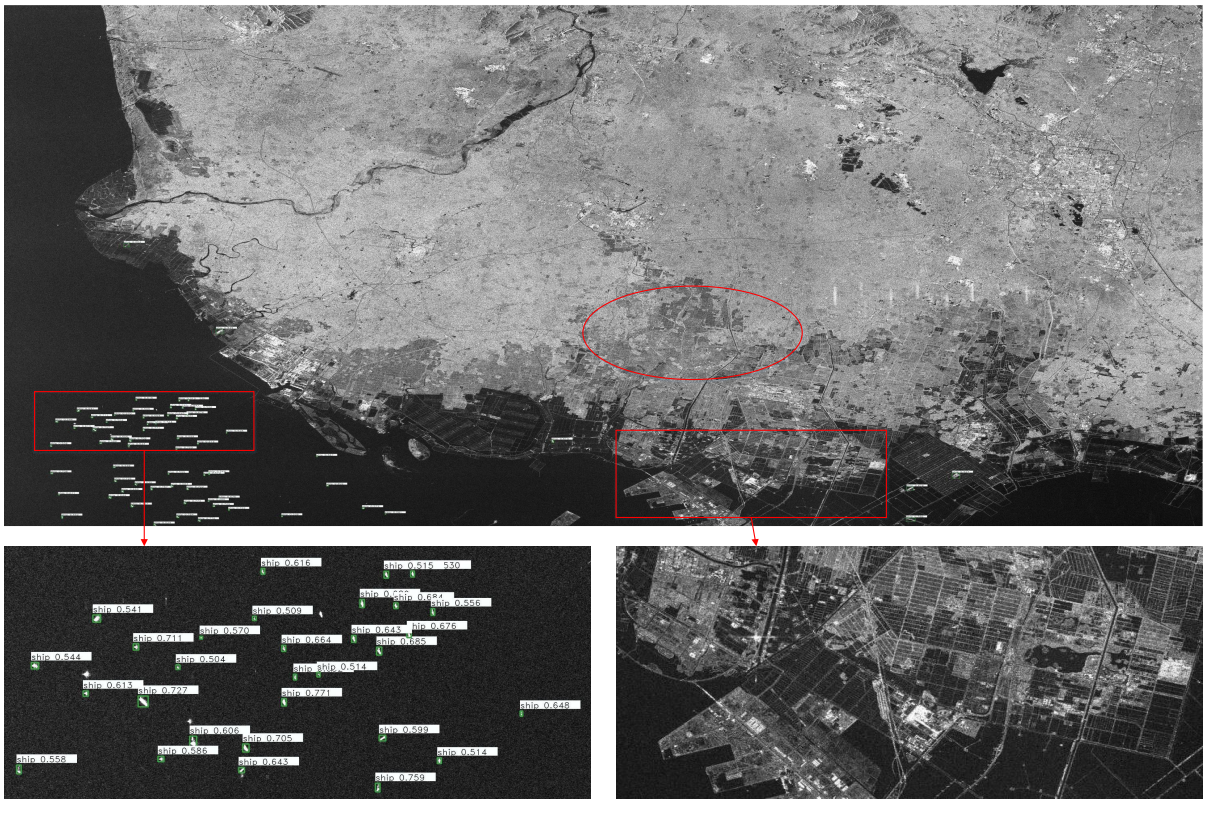
2) Detection: The method proposed in this article is comprehensively evaluated on the slice of Sentinel 1 and some results are shown in Table IV. In Table IV, Pre represents preclassification. In this section, the necessity of preclassification and the effectiveness of SSE are verified. The results show that when preclassification and SSE are joined with the detector, the precision can reach 90.6%. When detecting the objects without SSE and preclassification, too many false positives appeared, resulting in low detection efficiency. The proposed method achieves the superior results on performance and speed.

From Table IV, it can be seen that the proposed method is efficient for ship detection in large-scale SAR images.

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(a)



(b)

Fig. 10. Results in large-scale SAR images. (a) is the result without SSE and preclassification. There are many false alarms in nearshore and inland. (b) is the result with SSE and preclassification.

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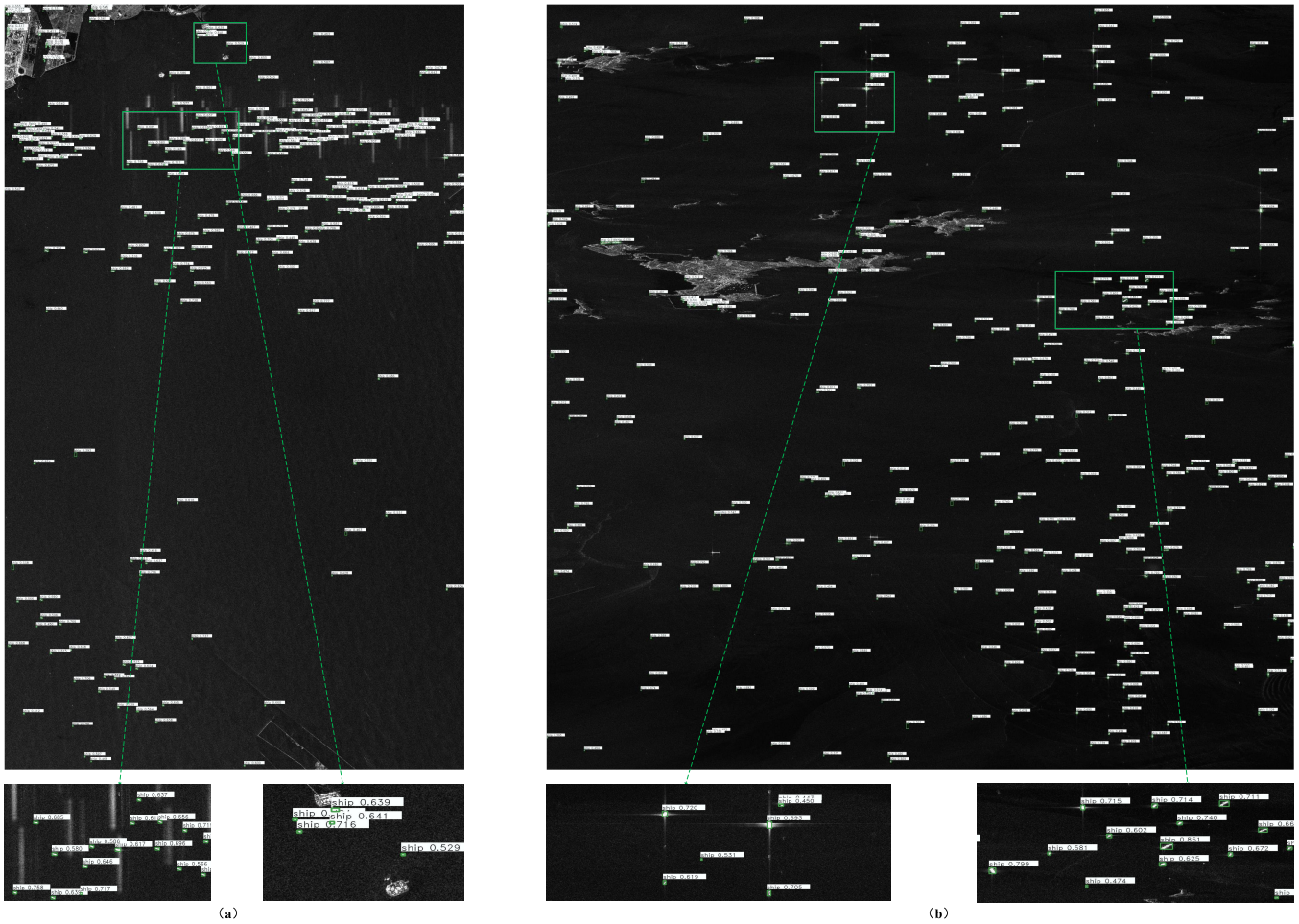


Fig. 11. Some other ship detection results in large-scale SAR images.

TABLE IV
EFFECT OF PRECLASSIFICATION AND SSE ON LARGE-SCALE SAR IMAGES

Method	Pre	TP	FP	Recall	Precision	Time Cost (s)
CenterNet	×	66	36	84.6%	64.7%	39.2
SSE-CenterNet	√	58	6	74.4%	**90.6**%	32.6

TABLE V
COMPARISON WITH OTHER METHODS IN LARGE-SCALE SAR IMAGES

Method	TP	FP	Recall	Precision
SSD-300	51	13	65.4%	79.7%
YOLOv3	48	15	61.5%	76.2%
the proposed method	58	6	74.4%	**90.6**%

DLA enables the inference time of the network less than other large models, such as ResNet-101, and extracts powerful features for object detection. Although the real-time detection of large-scale SAR images still has a long way to go, the proposed method has made some contributions.

To verify the performance of our method in large-scale SAR images, the method proposed in this article is compared with other ship target detection methods. As shown in Table V, the proposed method obtains the highest precision. It can be seen that the method in this article has better detection performance. At the same time, the proposed method has a better performance for shorelines or other more complicated scenes, and the method further reduced false positives. As shown in Fig. 10, [Fig. 10(a)] is the result without the SSE attention module and preclassification. First, it can be seen that there are many false alarms inshore. Many bright areas in the inshore area are similar to ship targets, thus causing severe false alarms. This problem has always existed in the development of ship target detection. Secondly, the residential areas in the inland area also caused some interference. For ship targets, the detection of inland areas is a waste of computing resources. The enlarged red rectangle on the left in [Fig. 10(a)] indicates that it has good performance in offshore. However, the red rectangle on the right shows that it has many false alarms inshore and cannot handle interference information well. [Fig. 10(b)] is the detection result with SSE attention module and preclassification. Without reducing the detection rate, false positives are significantly reduced. It is seen from the red oval box in Fig. 10 that preclassification is necessary and critical. Preclassification can not only reduce the calculation and improve the detection speed, but also effectively reduce the false alarms in the inland area.

Some other ship detection results in large-scale SAR images are shown in Fig. 11. As can be seen from Fig. 11, the method

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proposed in this article can effectively reduce false alarms. All the above experimental results show that, for ship detection in large scene SAR images, the method proposed in this article has advantages in both detection performance and detection efficiency.

IV. Conclusion

A detection method based on CenterNet with SSE attention module is proposed in this article, to solve the ship detection problem in large-scale SAR images. CenterNet is different from the anchor-based methods in that it locates the center point of the target through key point estimation, which can avoid missing detection of ship target in large-scale SAR image effectively. Moreover, the SSE attention module is introduced into CenterNet, which can simultaneously integrate attention in channels and spaces. With SSE, the CenterNet can integrate more advanced semantic features, to reduce the false alarms caused by inshore and inland interference.

Experimental results on SAR-ship-data set show that, the mAP of ship detection in large-scale SAR images via the proposed method reaches 94.7%, which can prove that the proposed method achieves a better performance than other state-of-the-art ship detection methods, such as faster R-CNN, SSD, YOLO, FPN, and their variations. But it also needs to be noted that, although the proposed method has better detection performance and can effectively reduce false alarms, it cannot completely eliminate all false alarms for complex background, which requires further analysis and research.

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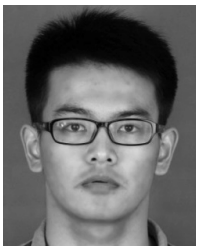
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